Homework Assignment 3

Show all your computations and justify your answers.

1. **(20 points)** Find the numbers at which f is discontinuous. At which of these numbers is f continuous from the right, from the left, or neither? Sketch the graph of f.

(a) $$f(x) = \begin{cases} e^x & \text{if} \quad x \le 0, \\ -(x-1)^2+2 & \text{if} \quad 0 < x < 2, \\ \ln(x-1)+2 & \text{if} \quad x \ge 2. \end{cases}$$

(b) $$f(x) = \begin{cases} \sin^2(2x) & \text{if} \quad x \le -\frac{\pi}{2}, \\ \frac{4x^3-4x^2-\pi^2x+\pi^2}{x-1} & \text{if} \quad -\frac{\pi}{2} < x < \frac{\pi}{2}, \\ \tan\left(x-\frac{\pi}{2}\right) & \text{if} \quad x \ge \frac{\pi}{2}. \end{cases}$$

2. **(20 points)** Find the values of a and b that make f continuous everywhere.

$$f(x) = \begin{cases} \frac{x^3-x^2-x+1}{x-1} & \text{if} \quad x < 1, \\ ax^2+bx+1 & \text{if} \quad 1 \le x < \pi, \\ 2\sin\left(x-\frac{5\pi}{2}\right) & \text{if} \quad x \ge \pi. \end{cases}$$

3. **(24 points)** Find the vertical and horizontal asymptotes of the following curves.

(a) $f(x) = \frac{x-1}{x^2+x-1}$. (b) $g(x) = \frac{\cos^2 x}{x^2+1} + 1$. (c) $h(x) = \frac{1+e^{2x}}{1+4e^{2x}}$.

4. **(16 points)**

(a) For $x > 0$, prove that the derivative of $f(x) = \sqrt{x}$ is $f'(x) = \frac{1}{2\sqrt{x}}$ by applying the definition of derivative as a limit.

(b) Explain why $f'(0)$ does not exist.

5. **(20 points)** Find a, b, c and d in $\mathbb{R}$ such that the following function is differentiable.

$$f(x) = \begin{cases} \sin x & \text{if} \quad x \le 0, \\ ax^3+bx^2+cx+d & \text{if} \quad 0 < x < 1, \\ -x^2+2x & \text{if} \quad x \ge 1. \end{cases}$$

Is f' differentiable?